

Trott and Bergen’s Hierarchical Linear Models: A Cross-Classified Mixed Effects Analysis

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Based on: Trott, S., & Bergen, B. (2023). Word meaning is both categorical and continuous.

Psychological Review, 130(5), 1239–1261. <https://doi.org/10.1037/rev0000420>

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1. Article Overview

Trott and Bergen (2023) tackle a longstanding debate in psycholinguistics: how do humans mentally represent words with multiple meanings? Two competing frameworks exist. *Categorical* (or “Mental Dictionary”) theories hold that the mind stores discrete, dictionary-like entries for distinct word senses. *Continuous* (exemplar-based) theories treat meaning as trajectories through a continuous semantic space, with no hard boundaries between senses. Trott and Bergen argue that neither account alone is sufficient. They propose two *hybrid* theories: “hybrid meaning theory” incorporates both discrete sense boundaries and continuous semantic distances while “hybrid+ theory” claims that the mind further differentiates between homonymy and polysemy in this hybrid space.

Trott and Bergen’s research question was: Do continuous semantic similarity and dictionary-derived categorical boundaries independently predict cognitive processing difficulty in a primed sensibility judgment task involving lexically ambiguous terms as measured by response times and accuracy?

In two behavioral experiments, Trott and Bergen used a sensibility judgment paradigm, using dictionaries to identify sense boundaries. In each trial, participants were shown two sentences on consecutive screens containing the same critical word (a polyseme or homonym)¹. The critical word in the first (priming) sentence might or might not have the same sense as the same critical word in the second (target) sentence. They also created a number of sentence pairs with unambiguous filler words, each matched for the lexical characteristics of one of the critical wordforms. In approximately 50% of the sampled sentence pairs, both sentences were sensible. About $\frac{1}{3}$ of the pairs contained one nonsensical sentence, and in $\frac{1}{6}$ of the pairs, both sentences were nonsensical. Participants were tasked with judging whether

¹**Homonymy:** A single word form possessing multiple *unrelated* meanings that do not share conceptual features (e.g., bank as a financial institution vs. the side of a river). In standard lexicographic practice, homonymous senses are listed as separate headword entries. **Polysemy:** A single word form possessing multiple *related* meanings (e.g., paper as a material, as a document, as an academic assignment, or as a newspaper). In standard lexicographic practice, polysemous senses are listed as definitions under the same headword entry.

each sentence (independently) “makes sense,” with response times revealing the degree of semantic facilitation or interference between prime–target pairs. The participant only saw, and judged the sensibility of, one sentence at a time. Participants’ accuracy in their sensibility judgments was also recorded (but was less informative than analysis of response times).

Each participant saw a subset of possible stimuli. The experimental design was repeated measures within participant. In both experiments, Trott and Bergen presented each participant with 56 critical (prime–target) sentence pairs interspersed with 56 filler sentence pairs, i.e. 112 sentence pairs. Each participant thus saw and evaluated the sensibility of 224 sentences in a session, but only the target sentence of critical pairs (not filler pairs) was included in the analysis. Thus, 56 trials (sentences) per session (participant) were used in the data analysis.

The results supported their “hybrid meaning theory.” They found that response time on the target trial was systematically related to the continuous distance between the prime and target contexts, above and beyond the existence of a sense boundary between those contexts. This result is inconsistent with the predictions of theories falling under the Mental Dictionary Framework. Both response time and accuracy were independently modulated by the existence of a sense boundary, which is inconsistent with the predictions of pure exemplar (i.e., continuous) theory for this task. They also found no evidence that the size of this effect depended on the kind of ambiguity (i.e., homonymy vs. polysemy), suggesting that these phenomena do not elicit categorically distinct behavior on the task. Altogether, this suggests that *hybrid meaning theory* accounts best for the behavioral signatures they measured of how humans represent the meaning of ambiguous words.

2. Experimental Variables

- **BERT cosine distance** (`distance_bert`) is a quantitative measure of semantic distance (i.e., dissimilarity) between the prime and target sentence contexts.² Cosine

²Semantic distance is operationalized as the cosine distance between BERT-derived contextualized word embeddings. Cosine distance can be defined as $(1 - \text{cosine similarity})$, where cosine similarity has a range of $[-1, 1]$. The range of cosine distance is thus $[0, 2]$, where two proportional vectors have a cosine distance of 0,

distance values of word pairs range from 0 (identical meanings) to 1 (no relatedness) to 2 (opposite meanings).

- **Same sense** (`same`) indicates whether the two meanings expressed by a word were categorized by lexicographic experts (in two dictionaries) as same or different senses.
- **Ambiguity type** (`ambiguity_type`) is the classification of an ambiguous word as a polysemous or homonymous word. Trott and Bergen used both dictionaries to annotate whether different-sense items were classified by lexicographers as related via homonymy or polysemy; meanings listed as separate entries were annotated as homonymy, and those listed in the same entry were annotated as polysemy.
- **Average sentence-priming RT** (`prior_rt`) is a control variable. In their second experiment, the primary mixed-effects models controlled for sentence-specific baseline difficulty using the average response time (across all participants) of the target sentence when it appeared as a priming sentence, rather than relying primarily on lexical statistics of the target word such as frequency, concreteness, and length.
- **Response time** (`log_rt`), an outcome variable, is an indicator of cognitive processing difficulty. Response times (RT) were recorded in milliseconds. For analysis, RT were transformed by taking their (natural) logarithm due to their skewed distribution.

3. Data Structure

In Trott and Bergen’s two experiments, participants judged the sensibility of a presented sentence that included a word with multiple meanings (i.e., a polyseme or homonym) or an unambiguous “filler word.” Two consecutively presented sentences constituted a priming-target sentence pair.

The unit of analysis for their study was a single trial which includes a participant being presented, and judging the sensibility of, a target sentence. The relationship (or comparison)

orthogonal vectors have a distance of 1, and opposite vectors have a distance of 2.

between the ambiguous word’s use in the prime and target sentences is recorded with each trial (via the variables `distance_bert`, `same`, and `ambiguity_type`). The priming sentences themselves were not a unit of analysis (their role was solely to establish a semantic context whose relationship to the target sentence is captured by the sentence-pair-level predictors `distance_bert` and `same`); however, each participant’s response time to a sentence presented as the prime was averaged across all participants and used as a control variable (`prior_rt`) in analyzing instances of that sentence presented as the target sentence.

Terminological Notes

Four terms (word, word sense, stimulus sentence, and target sentence) recur throughout this discussion and merit explicit distinction, because they operate at different levels of the stimulus hierarchy and are easily conflated or confused.

Word (wordform, item). The ambiguous lexical item itself—e.g., *lamb*, *bank*, *cell*—referring to the full semantic range associated with the lexical form and not any particular sense that emerges in a sentence context. Trott and Bergen’s final stimulus set contained 104 critical words (30 homonyms, 74 polysemes). This is the unit that indexes the word-level random intercept ($1 \mid \text{word}$) in their `lme4` model, as described further below.

Word sense. One of the two primary meanings associated with a critical wordform, as determined by dictionary classification. Each critical word has exactly two primary senses (e.g., *lamb* as an animal vs. as a food-sense). Trott and Bergen created two sentences per sense, yielding four stimulus sentences per critical word and six possible sentence pairs (two same-sense, four different-sense). Word sense is not itself a grouping variable in the statistical model; rather, it is encoded analytically through the fixed-effect predictors `same` (whether prime and target instantiate the same dictionary sense) and `ambiguity_type` (polysemy vs. homonymy, a word-level property).

Stimulus sentence. Any sentence presented to a participant during the experiment, regardless of its role or whether it contributed directly to the analysis. This set includes all (a) priming sentences, (b) target sentences, and (c) filler sentences. In the experiment, participants judge the sensibility of each of these sentences, and the response time is the outcome variable (`log_rt`).

Target sentence. The second sentence in a critical prime–target pair. Each target sentence presents the critical word in a specific sense context; the sentence-pair-level predictors (`distance_bert`, `same`) describe the *relationship* between that target sentence and the preceding prime sentence. Although all stimulus sentences provide dependent variable values, only those associated with target sentences are included in the data analysis.

Why the word-level random intercept is not per word sense or per target sentence. The random intercept (`1 | word`) accounts for the fact that each critical *wordform* was seen by multiple participants across separate experimental sessions: multiple trials share the same word k , violating the independence assumption of OLS. It captures baseline variability in response time attributable to the word itself—how inherently familiar, frequent, or ambiguous a given wordform is—pooling across all senses and all sentence contexts in which it appeared.

The intercept is not per *word sense* because sense-level information is already represented in the fixed effects. `same` and `ambiguity_type` capture the sense-relational structure between each prime–target pair; a separate random intercept at the sense level would be redundant and is not identified given that each participant saw each word only once (precluding within-participant, within-word comparisons across senses).

The intercept is not per *target sentence* because Trott and Bergen did not treat individual sentences as independent random-effect levels. Each critical word has up to four associated sentences, but what makes repeated observations non-independent is the

shared wordform—not the shared sentence frame. Modeling at the sentence level would require a finer-grained random-effects structure than Trott and Bergen’s design, in which each participant sees at most one sentence per word, does not support.

Average sentence-priming RT. Finally, note that `prior_rt`—the covariate derived from the average response time to a given sentence when it appeared in the priming (first) position—is a property of the specific *target sentence*, not of the word in the abstract. Because each critical word has four associated context sentences (two per dictionary sense), any of which may appear as a target sentence, different instantiations of the same wordform can carry different `prior_rt` values; the covariate therefore varies at the sentence level, not purely at the word level.

Despite this extended clarification of terms, one place where *I am blurring* these distinctions is the notation $(\text{prior_rt})_k$ in the formal model. As just stated, `prior_rt` is a target-sentence-level covariate (serving as a proxy for intrinsic sentence-processing difficulty), but the k subscript is a word-level index. Because each participant saw each word at most once, the target sentence presented and the word presented are in one-to-one correspondence within any given trial, so the subscript is unambiguous in context even if imprecise in the abstract. (Trott and Bergen also use this compression, but not explicitly.)

In Trott and Bergen’s design, the ambiguous word in each critical sentence pair was seen only once by a participant, but may have been seen by several participants in their own experiment sessions. Consequentially, each participant saw 56 unique critical words in their session (as well as 56 unique filler words that were not part of the analysis).

The experimental design had the following implicit rules (see Figure 1 as an illustration):

1. Each trial involved one participant and one stimulus (target) word.
2. A participant was linked to several trials, but a trial belonged to only one participant.
3. A stimulus word could be linked to several trials only if each trial was linked to a different participant.

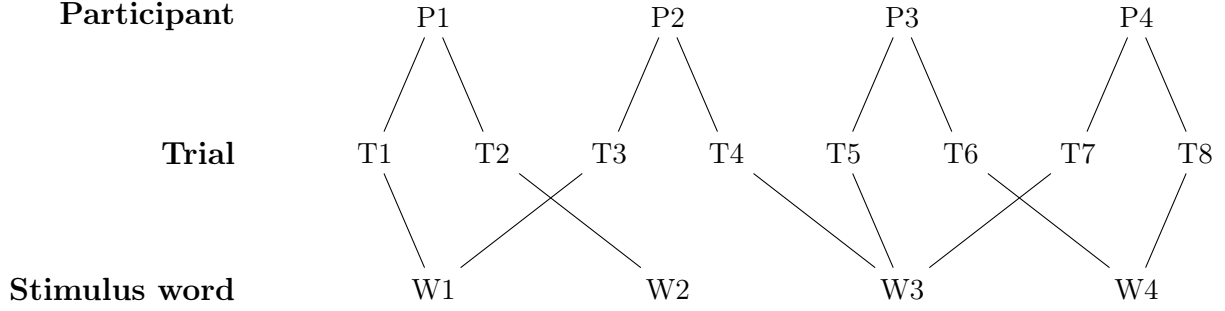


Figure 1: Illustrative crossed data structure for Trott and Bergen (2023). Trials are cross-classified by participant and stimulus word: each trial belongs to one participant and presents one stimulus word, while each participant contributes multiple trials and each stimulus word appears across multiple participants.

The data from this study, due to its repeated-measurements (or within-subjects) design, was analyzed using hierarchical linear models. The random effects structure was:

- $(1 + \text{distance_bert} + \text{ambiguity_type} + \text{same} \mid \text{ppt})$: Participant-level variation in response times. Random intercept: participant-specific deviation from grand mean $\log_rt$. Random slopes: participants’ response time sensitivity varied with changes in these predictors.
- $(1 \mid \text{word})$: Word-level variation in response times. Random intercept: some words are inherently more ambiguous or difficult to process.

4. Conceptual Foundation: Why Cross-Classified?

In a strictly nested design (the default in Raudenbush and Bryk’s [2002] earlier chapters), each lower-level unit belongs to one and only one higher-level unit — students nested within schools, measurements nested within patients. Trott and Bergen’s design departs from this in a critical way: *participants and (target) words constitute two fully crossed, independent grouping factors*. Any given participant saw a subset of items, and any given item was seen by a subset of participants, but no participant “belongs” to a specific word and no word “belongs” to a specific participant. As Raudenbush and Bryk (2002) put it in Chapter 12: “A cross-classification arises when lower-level units (e.g., students) share memberships in a unit

of one factor (e.g., a neighborhood) and can belong to different units of a second factor (e.g., different schools)” (p. 396).

In Trott and Bergen’s design, a single observation (trial) is the lower-level unit, cross-classified by *participant* (row factor, analogous to Raudenbush and Bryk’s “neighborhoods”) and *word* (column factor, analogous to “schools”). Because each participant saw each word at most once (one trial per participant–word combination), *every cell in the participant $\times$ word matrix contains at most one observation* — there is no “within-cell” sample from which to estimate a cell-specific residual above and beyond the *participant* and *word* random effects. This justifies the dropping of the within-cell index i that appears in Raudenbush and Bryk’s general notation.

5. Raudenbush and Bryk’s Framework for Cross-Classified Models

Raudenbush and Bryk’s general Level-1 (within-cell) model (p. 396) is:

$$Y_{ijk} = \pi_{0jk} + \sum_{p=1}^P \pi_{pj k} a_{pijk} + e_{ijk} \quad (1)$$

where case i is nested within row j (e.g., neighborhoods) and column k (e.g., schools) of the two-way array of observations, and $e_{ijk} \sim \mathcal{N}(0, \sigma^2)$.

The Level-2 (between-cell) model for each coefficient $\pi_{pj k}$ is:

$$\pi_{pj k} = \theta_p + \sum_{q=1}^Q (\beta_{pq} + b_{pqj}) X_{pqk} + \sum_{r=1}^R (\gamma_{pr} + c_{prk}) W_{prj} + b_{p0j} + c_{p0k} + d_{pj k} \quad (2)$$

where b_{p0j} is the residual random effect of row j , c_{p0k} is the residual random effect of column k , and $d_{pj k}$ is a row-by-column interaction random effect. When cell sizes are too small to estimate $d_{pj k}$ (and especially when $n_{jk} \leq 1$), the interaction term is dropped.

In Trott and Bergen’s application, the “level-2 row predictors” (W_{prj} , properties of participants) and “level-2 column predictors” (X_{pqk} , properties of [target] stimulus words) map as follows:

- **Row (participant) predictors:** There are none explicitly — participants are not characterized by any measured covariate; their contribution is entirely through the random effects b_{0j} , b_{1j} , b_{2j} , b_{3j} .
- **Column (word) predictor:** `ambiguity_type` is a property of the target wordform k , not of the sentence pair. Its value is the same regardless of which sentence pair instantiates that word. It enters as a fixed effect with a participant-level random slope, but its variance is between-word rather than between-cell.
- **Sentence-pair-level predictors:** `distance_bert` and `same` are properties of the specific prime–target sentence pair and are defined at the cell level jk — their values are fixed by which two sentences were paired, independently of which participant is responding. In Raudenbush and Bryk’s framework, they enter as fixed effects with participant-level random variation in their slopes.
- **Target-sentence-level covariate:** `prior_rt` is a property of the specific sentence used as the target, not of the wordform in the abstract. Because each critical word has up to four target sentences, different instantiations of the same word can have different `prior_rt` values. It enters with a fixed coefficient and no random slope.

This is a somewhat unusual feature of psycholinguistic designs that Raudenbush and Bryk’s examples do not perfectly anticipate: the primary predictors are neither purely row-level (participant) nor purely column-level (word) covariates, but sentence-pair-level (`distance_bert` and `same`) as well as word-level (`ambiguity_type`) properties of the stimulus. The solution Trott and Bergen implement — treating these as fixed effects with by-participant random slopes and a random intercept for items — is a standard `lme4` practice for such designs and is consistent with the spirit of Raudenbush and Bryk’s framework.

6. Mapping Trott and Bergen’s Design to Raudenbush and Bryk’s Framework

Before writing the models, it is important to clarify the variable structure.

Level-1 (trial-level) variables — vary trial by trial:

- **distance_bert**: a **sentence-pair-level** predictor. Its value is determined by the specific prime–target sentence pairing in cell jk and does not depend on which participant is responding. It is not reducible to a word-level property: the same wordform yields different cosine distance values across its six possible sentence pairings.
- **same**: also a **sentence-pair-level** predictor. Like **distance_bert**, it describes a relationship between the prime and target sentences in cell jk , not a property of the word or the participant.
- **ambiguity_type**: a **word-level** predictor. Whether a word is a polyseme or homonym is a property of the wordform itself; its value is constant across all sentence pairs involving the same word. It is given a by-subject random slope because participants may differ in their sensitivity to ambiguity type, but its variance is entirely between-word.

Control variable:

- **prior_rt**: a **target-sentence-level** covariate. It is the average first-trial RT for a specific sentence when it appeared in the priming position, averaged across all participants who saw it there. Because a given word can appear in up to four different target sentences, this covariate varies at the sentence level, not purely at the word level. It is treated as fixed (no random slope), serving as a proxy for intrinsic sentence-processing difficulty.

Outcome:

- **log_rt**: natural log of response time (ms), for correct trials only.

Random grouping factors:

- `ppt` (row factor, j): participant random intercept + random slopes for `distance_bert`, `same`, and `ambiguity_type`
- `word` / `item` (column factor, k): random intercept only

7. Trott and Bergen’s Level-1 and Level-2 Models

Because each participant–word cell contains at most one observation, the index i is dropped throughout. The observation is indexed only by j (participant) and k (item/word). The Level-1 “within-cell” model collapses: it is simply the observation itself with its associated error, which is absorbed into the residual structure.

One way to present this data structure in Raudenbush and Bryk’s framework (without interactions) is as follows.

7.1 Level-1 (Trial-Level) Model

$$Y_{jk} = \pi_{0jk} + \pi_{1jk}(\text{distance_bert})_{jk} + \pi_{2jk}(\text{same})_{jk} + \pi_{3jk}(\text{ambiguity_type})_{jk} + \pi_4(\text{prior_rt})_k + e_{jk} \quad (3)$$

$$e_{jk} \sim \mathcal{N}(0, \sigma^2) \quad (4)$$

where:

- Y_{jk} is `log_rt` for the trial of participant j responding to word (wordform) k
- π_{0jk} is the cell intercept (expected `log_rt` for participant j on word k when all predictors are zero)
- $(\text{distance_bert})_{jk}$ is the BERT cosine distance for the sentence pair in cell jk
- $(\text{same})_{jk}$ is the sense boundary indicator for the sentence pair in cell jk
- $(\text{ambiguity_type})_{jk}$ is the polysemy/homonymy classification for the word in cell jk
- $(\text{prior_rt})_k$ controls for the baseline difficulty of the word k that is instantiated by

the specific target sentence in cell jk .³ It is treated as having a *fixed effect only* (no random slope), consistent with Trott and Bergen’s `lme4` code in R.

- σ^2 is the residual (within-cell) variance

7.2 Level-2 (Between-Cell / Cross-Classified) Model

The four Level-1 coefficients become outcomes at Level 2. The intercept π_{0jk} is decomposed into contributions from participants (rows) and words (columns). The slope coefficients π_{1jk} , π_{2jk} , π_{3jk} are given random variation over participants (but not over words — only the intercept varies randomly over words in Trott and Bergen’s specification).

$$\pi_{0jk} = \theta_0 + b_{0j} + c_{0k} \quad (5)$$

$$\pi_{1jk} = \theta_1 + b_{1j} \quad (6)$$

$$\pi_{2jk} = \theta_2 + b_{2j} \quad (7)$$

$$\pi_{3jk} = \theta_3 + b_{3j} \quad (8)$$

$$\pi_4 = \theta_4 \quad (9)$$

with distributional assumptions:

$$\begin{pmatrix} b_{0j} \\ b_{1j} \\ b_{2j} \\ b_{3j} \end{pmatrix} \sim \mathcal{N}(\mathbf{0}, \Sigma_b), \quad c_{0k} \sim \mathcal{N}(0, \tau_{c_{00}}) \quad (10)$$

³As explained earlier, `prior_rt` is a target-sentence-level covariate, but the k subscript is a word-level index. I chose this “compression” option instead of the notation $(\text{prior_rt})_{s(k)}$ where $s(k)$ denotes the specific target sentence instantiating word k in cell jk or $(\text{prior_rt})_{s:k}$ explicitly nesting sentence s within word k .

where:

Symbol	Interpretation
θ_0	Grand mean log RT (population intercept)
θ_1	Fixed effect of <code>distance_bert</code> on log RT
θ_2	Fixed effect of <code>same</code> (sense boundary) on log RT
θ_3	Fixed effect of <code>ambiguity_type</code> on log RT
θ_4	Fixed effect of <code>prior_rt</code> on log RT
b_{0j}	Random intercept deviation for participant j
b_{1j}	Random slope deviation for <code>distance_bert</code> for participant j
b_{2j}	Random slope deviation for <code>same</code> for participant j
b_{3j}	Random slope deviation for <code>ambiguity_type</code> for participant j
c_{0k}	Random intercept deviation for item/word k
Σ_b	4×4 unstructured covariance matrix for participant random effects
τ_{c00}	Variance of item random intercepts

The participant random effects $(b_{0j}, b_{1j}, b_{2j}, b_{3j})^\top$ are *correlated* with each other, corresponding to lme4's `(1 + distance_bert + same + ambiguity_type | ppt)` (Bates et al., 2003). The word random effect c_{0k} is *independent* of the participant effects and has only an intercept (no random slopes), corresponding to `(1 | word)`. This is exactly the structure in Trott and Bergen's R code⁴:

```
model_full_rt = lmer(data = df_merged_correct,
  log_rt ~ distance_bert + ambiguity_type * same + prior_rt +
    (1 + distance_bert + ambiguity_type + same | ppt) +
    (1 | word),
  control=lmerControl(optimizer="bobyqa"),
  REML = FALSE)
```

7.3 No Apparent Centering = Uninterpretable Intercept

It appears that Trott and Bergen used the “natural X metric” — no centering was applied to the continuous predictors before modeling. Because no centering was applied, the intercept θ_0

⁴Trott and Bergen use the variable label `subject` instead of `ppt` (participant)

represents the expected `log_rt` when all continuous predictors are simultaneously zero — i.e., when `distance_bert` = 0 (perfectly identical BERT embeddings), `prior_rt` = 0 (a mean first-trial log RT of 0, which corresponds to an RT of 1 ms), and the categorical predictors are at their reference levels (`same` = TRUE, `ambiguity_type` = Polysemy). A `prior_rt` of zero is entirely outside the range of the observed data, making the intercept uninterpretable in any substantive sense.

This may be a common and generally unproblematic choice in psycholinguistics when the research focus is entirely on the *slopes* (the fixed effects of the predictors), not the intercept. Trott and Bergen’s substantive claims — that `distance_bert` and `same` each independently predict log RT — depend only on the coefficients θ_1 and θ_2 , which are unaffected by centering.

8. Two Approaches to Test a Predictor

Trott and Bergen evaluated their hypotheses with likelihood-ratio χ^2 tests comparing nested models (in the psycholinguistic sense of *nested*).⁵ For example, they compared a model with fixed effects for sense boundary and cosine distance (`model_distance_same_rt`) to a model that omits (only the fixed effect of) sense boundary (`model_distance_rt`). The question is: how is this different from running the regression in HLM 8 (Raudenbush et al., 2019) and testing the significance of π_{2jk} to show that `same` is a statistically significant predictor of response time? It turns out that the two approaches are fundamentally different in what they test, how they use the data, and when each is valid.

⁵I suppose it is apt that, in a study of words with multiple meanings, the term *nested* carries distinct meanings. In hierarchical linear modeling, nesting refers to a property of the data structure: lower-level units of analysis are contained within higher-level units. In the psycholinguistic literature, by contrast, *nested models* refers to a property of competing statistical specifications: one model is nested within another when it can be obtained by constraining one or more parameters of the fuller model to zero. Goodness-of-fit comparisons between such model pairs are evaluated via the likelihood-ratio test which follows an approximate χ^2 distribution and allows for hypothesis testing of H_0 : the data fits the nested model vs. H_1 : the data fits the full model.

8.1 The Two Approaches

8.1.1 Approach 1: The t -ratio (Single-Parameter Test of a Fixed Effect)

Note first a subtlety: what we called “ π_{2jk} ” in Level-1 notation, for example, is a coefficient that varies across participants. What is *actually* being tested by the t -ratio is θ_2 —the fixed, population-mean value of that coefficient. The t -ratio tests whether that population mean is distinguishable from zero.

When we run a single HLM model in R or HLM 8 and inspect the coefficient table, every fixed effect estimate $\hat{\theta}_p$ comes with a standard error and a t -value:

$$t = \frac{\hat{\theta}_p}{SE(\hat{\theta}_p)} \quad (11)$$

This tests $H_0 : \theta_p = 0$, i.e., that the population-average effect of a predictor on `log_rt` is zero—using information from *a single fitted model*.⁶ In HLM 8, this t -ratio is *standard output* in the Fixed Effects Table for every model we run, under both REML and full ML estimation. In R—with the `lmerTest` extension package (Kuznetsova et al., 2026)—it appears as a t -value with Satterthwaite-approximated degrees of freedom.

8.1.2 Approach 2: The Likelihood-Ratio Test (LRT)

Trott and Bergen’s approach fits *two separate models*—e.g., one with `same` as a fixed effect (`model_distance_same_rt`) and one without it (`model_distance_rt`)—and compares their deviances D . If L_0 is the likelihood, given the observed data, of the simpler model (in this case `model_distance_rt`) and L_1 is the likelihood of the full model (in this case `model_distance_same_rt`), then:

$$\Delta D = D_0 - D_1 = (-2 \log L_0) - (-2 \log L_1) \sim \chi^2(m) \quad (12)$$

⁶Raudenbush and Bryk note (p. 58) that this ratio is asymptotically unit normal, but a t distribution with approximately $J - S_q - 1$ degrees of freedom (where J is the number of level-2 units and S_q is the number of level-2 predictors for that coefficient) is usually more accurate.

where m is the difference in the number of estimated parameters (here $m = 1$, since the only difference is the addition of θ_2). A large ΔD means the fuller model fits the data substantially better—so much better that omitting the predictor `same` constitutes a significant degradation in fit.

As Raudenbush and Bryk (2002) make explicit (p. 61), *the LRT for fixed effects is only valid under full maximum likelihood (MLF), not restricted maximum likelihood (MLR or REML)*. Under REML, the likelihood function is defined differently for models with different fixed effects, making their deviances non-comparable. This is why Trott and Bergen set `REML = FALSE` in every `lmer()` call.

8.2 Key Differences

1. Single model vs. two models. The t -ratio requires only one estimation run; the LRT requires two.⁷

2. What is being measured. The t -ratio asks: *given everything else in this model, is θ_2 distinguishable from zero?* The LRT asks: *does the model that includes `same` fit the data significantly better than the model that excludes it?* These are logically equivalent questions for a single predictor in large samples, but they answer slightly different things and can diverge in finite samples or when model complexity differs in more than one way.

3. The REML restriction. Raudenbush and Bryk’s Table 3.2 (p. 58) summarizes it clearly:

Type of hypothesis	Fixed effects
Single-parameter	t -ratio (MLF or REML)
Multi-parameter	general linear hypothesis (MLF or REML)
LRT for fixed effects	MLF only

The t -ratio is available regardless of whether you use REML or full ML. The LRT for fixed effects requires full ML. HLM 8 defaults to REML for two-level models.

⁷This is exactly Raudenbush and Bryk’s first listed advantage of the t -ratio / general linear hypothesis approach (p. 61): “any set of linear hypotheses can be tested after computing estimates for just one model.”

4. Single vs. multi-parameter hypotheses. For `same`, which has one degree of freedom (TRUE vs. FALSE), the t -ratio and the LRT are asymptotically equivalent—squaring the t -ratio gives approximately the $\chi^2(1)$ statistic. So for this specific predictor, the two approaches will lead to the same conclusion in practice. The distinction becomes more meaningful when testing multi-parameter hypotheses: for instance, if `ambiguity_type` had more than two levels, dropping it from the model would constitute a multi-parameter test, and the LRT would be more natural than inspecting multiple individual t -ratios.

8.3 Why Trott and Bergen Chose the LRT

Given that the t -ratio and LRT are asymptotically equivalent for single- df predictors, Trott and Bergen’s choice to use the LRT throughout is likely a matter of disciplinary convention. Model comparison via `anova()` on `lmer` objects has been widely adopted in psycholinguistics in the wake of Barr et al.’s (2013) mandate for maximal random effects structure—which Trott and Bergen cite.⁸ I speculate that Trott and Bergen chose the LRT to align with their theoretical framing; the LRT asks “does the model that includes the `same` predictor fit the data significantly better than the model that excludes it?”—a question that maps more cleanly onto their theoretical agenda than the t -ratio’s question of whether a coefficient differs from zero.

Both approaches are defensible. If we use HLM 8, the t -ratios in the Fixed Effects Table are the natural output and are appropriate for testing individual predictors. If we want to replicate Trott and Bergen’s exact approach in HLM 8 (which we do in Section 10), we will need to switch to full ML, run paired models, and manually compute ΔD .

⁸Even within psycholinguistics, however, there is an ongoing methodological debate about when the LRT is preferable to t -ratios with Satterthwaite-approximated degrees of freedom. The two approaches answer logically equivalent questions for single- df predictors in large samples, but neither has achieved the status of an unchallenged field-wide standard.

9. The Full Model and Model Comparisons

Trott and Bergen’s preregistered analysis of Experiment 2 proceeds through three nested model comparisons (again, in the psycholinguistic sense of *nested*). It is important to note that *all models share the same random effects structure; only the fixed effects change*. This will be essential information when we compare their results in R to replicated models in HLM 8.

9.1 Model 1: model_distance_rt (Baseline)

$$Y_{jk} = \theta_0 + \theta_1(\text{distance_bert})_{jk} + \theta_4(\text{prior_rt})_k + b_{0j} + b_{1j}(\text{distance_bert})_{jk} + c_{0k} + e_{jk}$$

As just noted, even though `same` and `ambiguity_type` are omitted from the *fixed* effects, their random slopes for `subject` are *retained* in the random effects structure — a key feature of Trott and Bergen’s maximal-model philosophy (following Barr et al., 2013). The full Level-2 decomposition still includes b_{2j} and b_{3j} in Σ_b .

9.2 Model 2: model_distance_same_rt (Adds Sense Boundary)

$$Y_{jk} = \theta_0 + \theta_1(\text{distance_bert})_{jk} + \theta_2(\text{same})_{jk} + \theta_4(\text{prior_rt})_k + [\text{random effects as above}] + e_{jk}$$

In Trott and Bergen’s comparison of `model_distance_same_rt` vs. `model_distance_rt` via LRT, $\chi^2(1) = 45.57$, $p < .001$. The fixed effect of `same` significantly improves fit $\rightarrow$ *sense boundaries matter above and beyond continuous distance*. This disconfirms pure exemplar (continuous) theory.

9.3 Model 3: model_full_rt (Adds Ambiguity Type and Interaction)

$$\begin{aligned} Y_{jk} = & \theta_0 + \theta_1(\text{distance_bert})_{jk} + \theta_2(\text{same})_{jk} + \theta_3(\text{ambiguity_type})_{jk} \\ & + \theta_{23}(\text{ambiguity_type} \times \text{same})_{jk} + \theta_4(\text{prior_rt})_k + [\text{random effects}] + e_{jk} \end{aligned}$$

where θ_{23} is the fixed effect of the `ambiguity_type` $\times$ `same` interaction.

The combined `model_full_rt` (in Experiment 2) can be decomposed as follows:

$$Y_{jk} = \underbrace{\theta_0 + \theta_1 \text{db}_{jk} + \theta_2 \text{same}_{jk} + \theta_3 \text{at}_{jk} + \theta_{23} \text{at}_{jk} \cdot \text{same}_{jk} + \theta_4 \text{pr}_k}_{\text{Fixed part}} + \underbrace{b_{0j} + b_{1j} \text{db}_{jk} + b_{2j} \text{same}_{jk} + b_{3j} \text{at}_{jk} + c_{0k}}_{\text{Random part}} + e_{jk}$$

where `db` = `distance_bert`, `at` = `ambiguity_type`, `pr` = `prior_rt` for readability. Note that the interaction product (`ambiguity_type` $\times$ `same`)_{*jk*} has no random slope; it enters only as a fixed effect.

9.4 Model 4: `model_interaction_no_distance_rt` (Removes `distance_bert` from Model 3)

In comparison of `model_full_rt` vs. `model_interaction_no_distance_rt` via LRT, $\chi^2(1) = 39.64$, $p < .001$. Adding `distance_bert` significantly improves fit $\rightarrow$ *continuous gradations predict RT above and beyond categorical sense information*. This disconfirms the Mental Dictionary Framework.

9.5 Model 5: `model_all_main_effects_rt` (Removes Interaction from Model 3)

In comparison of `model_full_rt` vs. `model_all_main_effects_rt` via LRT, $p > .2$. The interaction does not improve fit $\rightarrow$ *there is not sufficient evidence that homonymy and polysemy produce different boundary-crossing effects*. This is consistent with Trott and Bergen’s hybrid meaning theory, but not their hybrid+ theory which included a distinction between polysemous and homonymous effects.

10. Replicating Trott and Bergen’s Results Using the HLM Software

As shown above, Trott and Bergen (2023) used the `lmer` function from the `lme4` package (Bates et al., 2003) in R to fit their hierarchical linear models (when the dependent variable

was `log_rt`). I start with the experimental data in `polysemy_s2_main.csv`, and I will describe the steps I took to replicate Trott and Bergen’s preregistered analysis for experiment 2 (`exp2_prereg.Rmd`) *using the HLM software*. Along the way, I will point out what the output from the HLM software is telling me (relevant to Trott and Bergen’s study), and how it is different from the results of using the `lme4` package in R.

The most important preliminary decision is choosing the appropriate module in the HLM software. Because Trott and Bergen’s models are cross-classified, HCM2 is required. As has been discussed above, individual trials (level 1) are simultaneously crossed by two grouping factors — subjects ($N = 187$) and words ($N = 104$). A given participant responds to many words, and a given word is seen by many participants, but no participant is nested within any word or vice versa. Participants will be the row factor (larger N) and words will be the column factor (smaller N), following the HLM guide’s convention.

10.1 Phase 1: Data Preparation in R (Before Using HLM)

I must first perform Trott and Bergen’s data cleaning steps in R to create clean CSV files for import into HLM 8.

Step 1.1 — Reproduce their filtering pipeline

Working from `polysemy_s2_main.csv`, I apply the exact steps from `exp2_prereg.Rmd`.⁹

Step 1.2 — Dummy-code categorical predictors

HLM software expects numeric predictors. Trott and Bergen’s `ambiguity_type` (“Polysemy” vs. “Homonymy”) and `same` (TRUE vs. FALSE) must be explicitly dummy-coded.

⁹Reproducing their filtering pipeline includes:

1. Filter to critical trials (no filler words, no practice trials).
2. Filter sentence pairs into first and second sentences (should be 56 per participant).
3. Recode version information to omit order (e.g., `"M1_a_M1_b" = "M1_b_M1_a"`), so it can later be merged with distance information.
4. Remove subjects who did not pass the bot check, remove trials with an $RT < 500\text{ms}$, `mutate(log_rt = log(rt))`, remove trials with an $RT > 3$ SDs above a subject’s mean, and remove subjects for whom more than half the observations have been removed (for slow or very fast RTs).
5. Among first trials, find `prior_rt = mean(log_rt)` for each sentence among all participants. Merge these results with those from previous steps.
6. Merge the experimental results with distance information and norming (relatedness) data.

Using Polysemy = 0 and TRUE = 0 (as their `lme4` below output reflects) preserves the reference levels.

```
df_merged$ambiguity_type = factor(df_merged$ambiguity_type, levels = c('Polysemy',
  ↪ 'Homonymy'))
df_merged$same = factor(df_merged$same, levels = c(TRUE, FALSE))

## Fixed effects in lme4 results include:
## ambiguity_typeHomonymy
## sameFALSE
## ambiguity_typeHomonymy:sameFALSE
```

The conversions of Trott and Bergen’s variable labels in HLM 8 are thus:

1. `log_rt` → `LOG_RT`
2. `distance_bert` → `DIST_BERT`
3. `ambiguity_type` → `HOMONYMY`
4. `same` → `DIFF_SENSE`
5. $(\text{ambiguity_type} \times \text{same}) \rightarrow \text{HOM_X_DIFF}$
6. `prior_rt` → `PRIOR_RT`

Step 1.3 — No centering

Trott and Bergen use the “natural X metric” for all continuous predictors (`distance_bert`, `prior_rt`). Thus, I do not center or standardize these variables before export.¹⁰

10.2 Phase 2: Preparing Data Files and Constructing the MDM File in HCM2

HCM2 requires three separate files corresponding to the three structural components of a cross-classified model.¹¹

- **File 1** — Level-1 (within-cell) file: Each row is one trial. This must include both grouping IDs plus all predictors and outcome variables.

– *File Name:* `hlm_l1_within_cell.dat`

¹⁰The `scale()` calls in their `.Rmd` scripts were only used for outlier removal on RT, not for transforming predictors going into the models.

¹¹Critical sorting requirement: The level-1 file must be sorted by row (`ppt`) ID, and within each `ppt`, by column (`word`) ID. The row-factor file must be sorted by `ppt` ID ascending; the column-factor file by `word` ID ascending.

- *Number of Variables*: 6 (the two IDs are declared separately as row ID and column ID, not counted in this box).
- *Data Format*: (2F4.0, 2F12.9, 3F3.0, F12.9) [See Table 1]
- **File 2** — Row-factor (ppt) file: One row per unique participant. Since Trott and Bergen have no level-2 predictors, this file needs only the participant ID and a dummy variable.
 - *File Name*: `hlm_row_subjects.dat`
 - *Number of variables*: 1 (dummy)
 - *Data Format*: (F4.0, F3.0)
- **File 3** — Column-factor (word/item) file: One row per word.
 - *File Name*: `hlm_col_words.dat`
 - *Number of variables*: 1 (dummy)
 - *Data Format*: (F4.0, F3.0)

Table 1: Data format for different fields in HLM 8

Field	Column	Width	Format	Meaning
subject_id	Row ID	4	F4.0	Integer 1–187
word_id	Column ID	4	F4.0	Integer 1–104
log_rt	Var 1	12	F12.9	e.g. 7.451822237
distance_bert	Var 2	12	F12.9	e.g. 0.167989969
homonymy	Var 3	3	F3.0	0 or 1
diff_sense	Var 4	3	F3.0	0 or 1
homonymy_x_diffsense	Var 5	3	F3.0	0 or 1
prior_rt	Var 6	12	F12.9	e.g. 7.158355275

10.3 Phase 3: Specifying the RT Models

Model 0 (`model_unconditional_rt`): The unconditional model contains no predictors at either level. Its sole purpose is to decompose the total variance in `LOG_RT` into three sources simultaneously attributable to the cross-classified structure:

$$\text{LOG_RT}_{ijk} = \theta_0 + b_{00j} + c_{00k} + e_{ijk} \quad (13)$$

Model 1 (model_full_rt): LOG_RT is selected as the Level 1 outcome variable. Each Level 1 predictor is added as an uncentered variable: DIST_BER, DIFF_SEN (= same), HOMONYMY (= ambiguity_type), HOM_X_DI, PRIOR_RT. Level 2 model is set to match Equations 5-9 in Section 7.2. Model saved as .hlm file. The number of (micro) iterations is set at 1000.

Models 2–5: The remaining four models differ only in which fixed effects are included. The random effects structure is the same (for applicable parameters).

An analysis is run for each model and output saved as .html files.

10.4 Phase 4: Reading the HCM2 Output for the Unconditional Model

Before we run our full analyses, it is recommended that we run an analysis on the unconditional model, i.e., no predictors included (See Equation 13). This enables us to partition the variance components and validate our two-level structure and cross-classification.

10.4.1 Fixed Effects

The estimated grand mean intercept was $\hat{\theta}_0 = 6.981$ ($p < .001$). Reverting this to the original time scale yields $e^{6.981} \approx 1.075$ seconds, representing the average response time across all participants and words prior to accounting for any predictors.

10.4.2 Variance Components

The three variance components partition the total outcome variance across the cross-classified grouping structure.

Source	Symbol	Variance	SD
Between participants (rows)	$\hat{\tau}_{b_{00}}$	0.04122	0.20302
Between words (columns)	$\hat{\tau}_{c_{00}}$	0.00749	0.08652
Within cell (residual)	$\hat{\sigma}^2$	0.09821	0.31339
Total		0.14692	

10.4.3 Intraclass Correlations

The intraclass correlations (ICCs) quantify how much of the total variance in `log_rt` is attributable to each grouping factor, and thereby justify the use of HCM2 over OLS regression.

The participant-level ICC is:

$$\rho_{\text{participants}} = \frac{\hat{\tau}_{b_{00}}}{\hat{\tau}_{b_{00}} + \hat{\tau}_{c_{00}} + \hat{\sigma}^2} = \frac{0.04122}{0.14692} = 0.281 \quad (14)$$

The word-level ICC is:

$$\rho_{\text{words}} = \frac{\hat{\tau}_{c_{00}}}{\hat{\tau}_{b_{00}} + \hat{\tau}_{c_{00}} + \hat{\sigma}^2} = \frac{0.00749}{0.14692} = 0.051 \quad (15)$$

These estimates indicate that 28.1% of the variance in `log_rt` is attributable to stable between-participant differences in response time, and an additional 5.1% to stable between-word differences in processing difficulty.¹² Together, 33.2% of total variance resides in the grouping structure. Both random effects were statistically significant (participants: $\chi^2(186) = 3744.87$, $p < .001$; words: $\chi^2(103) = 748.20$, $p < .001$), confirming that both grouping factors must be retained in all subsequent conditional models.

10.4.4 Baseline Statistics for Model Comparison

The unconditional model provides the baseline deviance against which all five preregistered conditional models can be compared (see Section 11):

- Deviance = 5183.800
- $k = 4$ ($\hat{\theta}_0, \hat{\tau}_{b_{00}}, \hat{\tau}_{c_{00}}, \hat{\sigma}^2$)
- AIC = Deviance + $2k = 5183.800 + 2(4) = 5191.800$

The reduction in deviance from this baseline to each conditional model reflects the combined explanatory contribution of the additional parameters.

¹²The word-level ICC (5.1%) is smaller than the participant-level ICC (28.1%), a pattern typical in psycholinguistics: individual differences in response time tend to exceed item-level difficulty differences when stimuli have been carefully normed, as in Trott and Bergen (2023).

10.5 Phase 5: Replication of Trott and Bergen’s Full Model in HLM 8

10.5.1 Full Model Fixed Effects

Fixed-effect estimates for `model_full_rt`, the most theoretically complete model, are presented in Table 2. All approximate degrees of freedom are 7,694.

Table 2: Fixed Effects for <code>model_full_rt</code>					
Parameter	Predictor	Coefficient	SE	<i>t</i>	<i>p</i>
$\hat{\theta}_0$	Intercept	3.064	0.190	16.09	< .001
$\hat{\theta}_1$	DIST_BERT	0.206	0.033	6.34	< .001
$\hat{\theta}_2$	HOMONYMY	0.025	0.017	1.46	.144
$\hat{\theta}_3$	DIFF_SENSE	0.059	0.010	6.17	< .001
$\hat{\theta}_4$	HOM_X_DIFF	0.016	0.016	0.97	.331
$\hat{\theta}_5$	PRIOR_RT	0.520	0.026	20.05	< .001

DIST_BERT ($\hat{\theta}_1 = 0.206$, $p < .001$) is a significant positive predictor of `log_rt`: larger cosine distance between prime and target sentence contexts is associated with slower response times (conditioned on the other predictors), consistent with the contribution of continuous distributional meaning to lexical processing difficulty. DIFF_SENSE ($\hat{\theta}_3 = 0.059$, $p < .001$) is also a significant positive predictor: different-sense trials elicited reliably slower RTs than same-sense trials after continuous cosine distance (along with the other predictors) was controlled, demonstrating a categorical sense boundary effect that cannot be reduced to distributional similarity. HOMONYMY ($\hat{\theta}_2 = 0.025$, $p = .144$) and the `ambiguity_type` $\times$ `same` interaction (HOM_X_DIFF; $\hat{\theta}_4 = 0.016$, $p = .331$) are not significant, indicating that ambiguity type did not independently predict RT and that the RT penalty for different-sense trials does not differ between polysemous and homonymous items. PRIOR_RT ($\hat{\theta}_5 = 0.520$, $p < .001$) is the strongest predictor in the model, confirming the importance of first-trial baseline difficulty as a covariate. *These results are consistent with the fixed-effect pattern reported by Trott and Bergen (2023).*

10.5.2 Variance Components for the Full Model

Variance components for `model_full_rt` are presented in Tables 3 and 4. The participant intercept variance was relatively large and highly significant, confirming substantial between-participant differences in baseline response time. The random slopes for `DIST_BERT` ($p > .500$) and `DIFF_SENSE` ($p > .500$) were not significantly different from zero, while the `HOMONYMY` slope variance was marginally significant ($p = .012$), suggesting modest but detectable variability across participants in sensitivity to ambiguity type. The word intercept variance declined from $\hat{\tau}_{c00} = 0.00749$ in the unconditional model to 0.00245, indicating that the fixed predictors — particularly `PRIOR_RT` and `DIST_BERT` — explain a substantial portion of between-word variation.

Table 3: Row-Factor (Participant) and Level-1 Variance Components: `model_full_rt`

Random Effect	Symbol	Variance	SD	$\chi^2(186)$	p
Participant intercept	$\hat{\tau}_{b00}$	0.04401	0.2098	943.21	< .001
<code>DIST_BERT</code> slope	$\hat{\tau}_{b11}$	0.00219	0.0468	166.07	> .500
<code>HOMONYMY</code> slope	$\hat{\tau}_{b22}$	0.00256	0.0506	231.88	.012
<code>DIFF_SENSE</code> slope	$\hat{\tau}_{b33}$	0.00021	0.0145	184.60	> .500
Level-1 residual	$\hat{\sigma}^2$	0.09201	0.3033		

Table 4: Column-Factor (Word) Variance Components: `model_full_rt`

Random Effect	Symbol	Variance	SD	$\chi^2(103)$	p
Word intercept	$\hat{\tau}_{c00}$	0.00245	0.0495	327.44	< .001

10.6 Discussion: Replicating Trott and Bergen’s Results in HLM 8

This section explained the process I used to replicate Trott and Bergen’s preregistered analysis for experiment 2 using the HLM software as well as the results. The replication exercise provided insight into the validity of the hierarchical linear model they used in support of hybrid meaning theory. At the level of fixed-effect conclusions, the two implementations agree closely. Both `DIST_BERT` and `DIFF_SENSE` emerge as significant positive predictors of `log_rt`, `HOMONYMY` and the `ambiguity_type` $\times$ `same` interaction are non-significant, and `PRIOR_RT` is

the strongest predictor in the model. The HCM2 coefficient estimates for `model_full_rt` (Table 2) are numerically consistent with Trott and Bergen’s `lme4` output, and the qualitative fixed-effect pattern — that both continuous distributional distance and categorical sense boundaries independently modulate processing difficulty, while ambiguity type does not — is reproduced in full. This provides a meaningful cross-implementation validation of Trott and Bergen’s primary theoretical claims.

11. HCM2 Replication Results Using the LRT Approach

In the previous section, it was shown that an analysis of Trott and Bergen’s data in HLM8 produced results that are consistent with the fixed-effect pattern reported in their presentation and evaluation of hybrid meaning theory. In this section, I’ll try something different. As noted previously, the approach used by HLM8 is the t-ratio for hypothesis testing, and the approach used by Trott and Bergen in R is the likelihood ratio test. So, here, I will essentially replicate Trott and Bergen’s LRT approach in HLM8. The motivation for this is to attain results that can be directly compared with those of `lme4`. *We will find that, taken together, the LRT results replicate the qualitative pattern reported by Trott and Bergen:* both the continuous distributional signal and the categorical sense boundary independently predict response time, while neither the main effect of ambiguity type nor its interaction with sense boundary reaches significance.

We will also discover, however, that the degrees of freedom of the χ^2 statistics reported in HLM8 differ from those reported by Trott and Bergen. This discrepancy originates in a fundamental architectural difference between `lme4` and HCM2 in how fixed and random effects are coupled. I will discuss two approaches that could in principle address this discrepancy and the one I would recommend for further replications.

11.1 Likelihood Ratio Tests

Three preregistered likelihood ratio tests following the model sequence specified in Trott and Bergen’s `exp2_prereg.Rmd` were conducted to evaluate the theoretical accounts of lexical ambiguity processing. Each test statistic was computed as the difference in deviance between a restricted and a fuller model (see Section 8.1.2). Results are presented in Table 5, alongside Trott and Bergen’s corresponding values for direct comparison. The degrees of freedom discrepancy is addressed in Section 11.2.

Table 5: Likelihood Ratio Tests: HCM2 Results and Trott and Bergen (2023) Comparison

Test	Comparison	HCM2		T&B (lme4)		<i>p</i>
		χ^2	<i>df</i>	χ^2	<i>df</i>	
<i>dis45_rt</i>	M4 vs. M5	49.23	4	45.57	1	< .001
<i>dis12_rt</i>	M1 vs. M2	41.47	5	39.64	1	< .001
<i>dis13_rt</i>	M1 vs. M3	0.93	1	~0.9	1	> .2, n.s.

M1 = `model_full_rt`; M2 = `model_interaction_no_dist`; M3 = `model_all_main_effects_rt`; M4 = `model_distance_same_rt`; M5 = `model_distance_rt`.

dis45_rt (disconfirming analysis, comparing M4 and M5 on RT) tested whether sense boundary information (`DIFF_SENSE`) improves model fit above cosine distance alone. The result was significant ($\chi^2 = 49.23$, $p < .001$), disconfirming a pure exemplar (distributional) account of lexical ambiguity processing.

dis12_rt tested whether cosine distance (`DIST_BERT`) improves fit above a model containing only categorical sense boundary and ambiguity type information. The result was significant ($\chi^2 = 41.47$, $p < .001$), disconfirming a pure Mental Dictionary account.

dis13_rt tested whether the `ambiguity_type` $\times$ `same` interaction improves fit above a model containing only main effects. The result was non-significant ($\chi^2(1) = 0.93$, $p > .2$), providing no evidence that the RT penalty for different-sense trials differs between polysemous and homonymous items. Trott and Bergen (2023) note that hybrid meaning theory is technically "compatible with a main effect of ambiguity type (i.e., an overall difference across homonymous and polysemous stimuli)" due to homonymous sense pairs tending to have larger BERT cosine distances than polysemous sense pairs (p. 8). Thus, once `distance_bert` is

controlled for, the type of boundary being crossed (polysemous vs. homonymous) should not be reflected in differences of cognitive processing difficulty. Finding no significant interaction is consistent with hybrid meaning theory but not hybrid+ theory. (This is the only comparison for which the HCM2 degrees of freedom exactly match Trott and Bergen’s.)

Taken together, *the LRT results in HLM8 replicate the qualitative pattern reported by Trott and Bergen (2023)*: both the continuous distributional signal and the categorical sense boundary independently predict response time, while neither the main effect of ambiguity type nor its interaction with sense boundary reaches significance. This pattern is consistent with their hybrid meaning theory, which posits that both sources of lexical information — discrete sense boundaries and graded semantic distance — independently modulate cognitive processing during lexical access.

11.2 Discrepancy Between lme4 and HLM 8 and Implications for Interpretation

11.2.1 Source of the Discrepancy

The degrees of freedom reported in Table 5 for *dis45_rt* and *dis12_rt* differ from those reported by Trott and Bergen. This discrepancy originates in a fundamental architectural difference between lme4 and HCM2 in how fixed and random effects are coupled.

Trott and Bergen specified their five models following a maximal random effects philosophy (Barr et al., 2013), in which the participant-level random effects structure — $(1 + \text{distance_bert} + \text{ambiguity_type} + \text{same} \mid \text{subject})$ — is held constant across all models regardless of which predictors appear in the fixed part. In lme4, the formula syntax permits this directly: the fixed and random components of the model are declared independently, and a predictor may have a random slope without appearing as a fixed effect. HCM2 does not support this separation.

As the HLM software guide (Section 1.1.2) makes explicit, each level-1 coefficient may take one of three forms: a fixed coefficient comprising only a grand mean θ_p with no random component; a non-randomly varying coefficient; or a randomly varying coefficient comprising

θ_p and a random effect b_{p0j} .¹³ There is no fourth option — a random effect without a corresponding fixed effect. The HLM for Windows interface enforces this mechanically: when a predictor is removed from the level-1 equation, its random slope is simultaneously dropped from the participant-level covariance matrix τ_b .

The consequence is that in the three models where Trott and Bergen omit a predictor from the fixed effects while retaining its random slope, the HCM2 runs produced a τ_b of reduced dimension (see Table 6).

Table 6: Random Effects Structure: Preregistered Specification vs. HCM2

Model	Target τ_b	HCM2 τ_b	Target k	HCM2 k
model_full_rt	4×4	4×4	18	18
model_all_main_effects_rt	4×4	4×4	17	17
model_distance_same_rt	4×4	3×3	16	12
model_interaction_no_dist	4×4	3×3	17	13
model_distance_rt	4×4	2×2	15	8

Because the LRT degrees of freedom equal the difference in k between the two models being compared, the reduced parameter counts produced $df = 4$ for *dis45_rt* ($k_{M4} - k_{M5} = 12 - 8$) and $df = 5$ for *dis12_rt* ($k_{M1} - k_{M2} = 18 - 13$) rather than the intended $df = 1$ for both. For *dis13_rt*, both models have the correct 4×4 τ_b and the degrees of freedom are exact.

11.2.2 Two Approaches to Address the Discrepancy

Two approaches could in principle address this discrepancy.

Approach A retains the five models as run and acknowledges the degrees of freedom inflation as a documented software constraint. The fixed effects in each model correspond to Trott and Bergen’s preregistered specification, and all three qualitative LRT conclusions are preserved. The χ^2 statistics are close to Trott and Bergen’s values (49.23 vs. 45.57 for *dis45_rt*; 41.47 vs. 39.64 for *dis12_rt*), and neither inflated degrees of freedom alters the significance: both tests would remain significant under any plausible df , since even

¹³Another possibility is a level-1 coefficient with *both* non-random and random sources of variation, but the HLM guide does not distinguish this as a fourth general form

$\chi^2(1) = 3.84$ ($\alpha = .05$) is far below the obtained values.

The main limitation of Approach A is that each LRT technically conflates the contribution of the added fixed effect with the contribution of the added random slope parameters, because the two models being compared differ in both their fixed and random structure. Strictly speaking, the test does not isolate the fixed effect alone.

Approach B re-runs the three affected models with the missing predictor(s) added back as both fixed and random effects, restoring the 4×4 τ_b in all five models. This produces $df = 1$ for *dis45_rt* and corrects the formal isolation of the fixed effect under test. However, Approach B introduces an insurmountable problem for *dis12_rt*: adding DIST_BERT back to *model_interaction_no_dist* as a fixed effect makes that model structurally identical to *model_full_rt* ($k = 18$ in both cases), collapsing the comparison to $df = 0$ and eliminating the test entirely. This is a more severe departure from Trott and Bergen’s analysis than the inflated degrees of freedom under Approach A, because it removes rather than adjusts one of the most theoretically important comparisons — whether continuous distributional distance predicts RT above and beyond categorical sense information.

11.2.3 Recommended Approach

I would recommend Approach A be adopted for any replication of Trott and Bergen (2023) that implements LRT in HCM2. The primary justification is that Approach B, while correcting the degrees of freedom for *dis45_rt*, fatally undermines *dis12_rt* — the test that directly bears on whether exemplar-based (continuous) representations contribute to lexical processing over and above categorical sense knowledge. Eliminating this comparison to gain a formally correct df in another would misrepresent the weight of evidence more severely than retaining inflated degrees of freedom with explicit documentation.

12. Conclusion

This critique has examined Trott and Bergen’s (2023) application of cross-classified hierarchical linear models to the question of lexical ambiguity representation from two complementary angles: a conceptual mapping of their design onto Raudenbush and Bryk’s (2002) framework for cross-classified models, and a direct replication of their Experiment 2 `log_rt` analysis in HLM 8 using the HCM2 module.

At the level of substantive conclusions, the replication strongly corroborates Trott and Bergen’s findings. Both `DIST_BERT` and `DIFF_SENSE` emerged as significant positive predictors of response time in HCM2, independently and above one another, while `HOMONYMY` and the `ambiguity_type` $\times$ `same` interaction were non-significant. All three preregistered likelihood ratio tests reproduced the qualitative pattern Trott and Bergen reported: sense boundaries predict RT above continuous distributional distance alone (*dis45_rt*: $\chi^2 = 49.23$, $p < .001$); continuous distance predicts RT above categorical sense information alone (*dis12_rt*: $\chi^2 = 41.47$, $p < .001$); and the interaction between ambiguity type and sense boundary does not improve fit (*dis13_rt*: $\chi^2(1) = 0.93$, $p > .2$). This cross-implementation convergence provides meaningful independent validation of Trott and Bergen’s hybrid meaning theory: the behavioral signatures of lexical ambiguity processing are not an artifact of the `lme4` implementation, but reflect a robust empirical pattern in the data.

The conceptual analysis, however, surfaces several features of Trott and Bergen’s design that merit explicit attention when situating their work within the Raudenbush and Bryk framework. First, their primary predictors — `distance_bert` and `same` — are sentence-pair-level properties of the stimulus, defined at the cell level jk and not reducible to either row-level (participant) or column-level (word) covariates in Raudenbush and Bryk’s sense. `ambiguity_type` is a word-level property, and `prior_rt` varies at the level of the specific target sentence, finer than the word-level subscript k used throughout. Raudenbush and Bryk’s Chapter 12 examples do not fully anticipate this stimulus hierarchy, making careful

terminological and notational work a prerequisite for any principled translation between the two frameworks. Second, Trott and Bergen’s use of the natural X metric leaves the intercept θ_0 uninterpretable in any substantive sense, though this does not affect the slope estimates on which their theoretical claims depend. Third, their choice of the LRT over the t -ratio is well-motivated by disciplinary convention and theoretical framing, but the two approaches are asymptotically equivalent for the single- df predictors tested here, and the distinction only becomes consequential for multi-parameter hypotheses — a point relevant to any extension of their design to predictors with more than two levels.

The replication also revealed a structural constraint of HCM2 that any practitioner should document when using it to reproduce `lme4` analyses. This is the fixed/random coupling constraint: HCM2 does not permit a predictor to carry a random slope without also appearing as a fixed effect, forcing a reduction in τ_b for three of the five preregistered models and inflating the degrees of freedom of the corresponding LRTs from $df = 1$ to $df = 4$ and $df = 5$. Adopting Approach A — retaining the models as run and evaluating the χ^2 statistics against the $\chi^2(1)$ critical value — is the appropriate response: the statistics of 49.23 and 41.47 vastly exceed any plausible threshold, and the alternative (Approach B) would eliminate a theoretically important comparison.

Taken together, these findings illustrate both the tractability and the limits of translating a psycholinguistic `lme4` analysis into the Raudenbush and Bryk framework. The cross-classified random effects logic of Chapter 12 maps cleanly onto Trott and Bergen’s design at the level of random intercepts and the variance decomposition, and the fixed-effect conclusions are fully reproducible in HCM2. The friction points — fixed/random coupling and sentence-level versus word-level predictor granularity — are software-architectural and notational rather than conceptual, and each has a principled resolution. For practitioners seeking to apply Raudenbush and Bryk’s framework to psycholinguistic designs of this type, the lesson is that the framework is adequate but translation requires explicit attention to predictor-level classification, estimation defaults, and the inferential machinery underlying model comparison.

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